Answer key: Lesson 6, Count Everything

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Count the squares by rows.

Answer: 10 squares

Each row has 5 squares. There are 2 rows. 5 and 5 make 10 squares.

Watch for: Counting a square twice.; Counting only one row.

S2. There are 3 jars. Each jar has 5 marbles.

Answer: 15 marbles

5 and 5 make 10. 10 and 5 make 15. There are 15 marbles in all.

Watch for: Writing 3 for the jars.; Counting only two jars.

S3. Each line joins two dots.

Answer: 5 lines

The lines are A to B, B to C, C to D, D to A and A to C. That is 5 lines.

Watch for: Counting the dots (4).; Counting a line twice.

S4. Every pair of numbers makes 9. Fill in the missing numbers.

Answer: 7; 6; 5

2 and 7 make 9. 3 and 6 make 9. 4 and 5 make 9. The list goes 1 and 8, 2 and 7, 3 and 6, 4 and 5.

Watch for: Writing 9 in every box.; Taking 2 away from 9 the wrong way round.

S5. Count the dots by rows.

Answer: 20 dots

Each row has 5 dots. There are 4 rows. 5 and 5 make 10, and 5 and 5 make 10. 10 and 10 make 20 dots.

Watch for: Counting only one row.; Counting a row twice.

S6. There are 9 squares. Some squares are yellow.

Answer: 5 squares

There are 4 yellow squares. 9 take away 4 is 5. So 5 squares are not yellow.

Watch for: Writing 4, the number of yellow squares.; Counting a square twice.

Think harder (page 3)

T1. Handshake Party. Four children, A, B, C and D, shake hands once with each other. A has already shaken hands with B, C and D.

Answer: 2; 1; 6

A made 3 handshakes. B has already shaken hands with A, so B makes 2 new ones, with C and with D. C has shaken hands with A and B, so C makes 1 new one, with D. D has none new. 3 + 2 + 1 = 6 handshakes.

Watch for: Writing 12 (4 times 3): every handshake is counted twice.; Counting B with A again.

T2. Mei has 4 tops and 2 pairs of shorts. Each top goes with each pair of shorts.

Answer: 8 outfits

Each of the 4 tops goes with 2 pairs of shorts. 2 and 2 and 2 and 2 make 8 outfits. There are 8 boxes.

Watch for: Writing 6 (4 and 2).; Writing 4, the number of tops.

T3. A big square is made of 4 small squares, like the yellow one.

Answer: 6; 2; 8

There are 2 rows of 3 small squares: 6 small squares. A big square covers 2 columns, and it can start in column 1 or column 2. So there are 2 big squares. 6 and 2 make 8 squares in all.

Watch for: Counting only the 6 small squares.; Counting the whole rectangle as a square.

T4. Six children have 3, 5, 6, 8, 9 and 11 stickers. Two children together have 14 stickers. Find every pair. Start with the child who has 3.

Answer: 11; 9; 8

3 and 11 make 14. 5 and 9 make 14. 6 and 8 make 14. The children with 8, 9 and 11 stickers are already in a pair, so these are all the pairs: 3 and 11, 5 and 9, 6 and 8.

Watch for: Missing one of the three pairs.; Counting 3 and 11 and 11 and 3 as two pairs.

Go further (page 4)

G1. Handshake Party. Children shake hands once with every other child. Think of A, B, C and so on. Use your paper people if you like.

Answer: 10; 15

5 children: A makes 4, B makes 3 new ones, C makes 2, D makes 1. 4 + 3 + 2 + 1 = 10. 6 children: 5 + 4 + 3 + 2 + 1 = 15.

Watch for: Writing 5 times 4 (20): every handshake is counted twice.; Stopping the list one number too soon.

G2. A grid is 3 squares across and 3 squares down. Count the squares of every size. The yellow one is a 2 by 2 square.

Answer: 9; 4; 1; 14

There are 9 small squares. A 2 by 2 square can start in 2 columns and 2 rows, so there are 2 and 2 which is 4. There is 1 big square, the whole grid. 9 + 4 + 1 = 14.

Watch for: Counting only the 9 small squares.; Forgetting the whole grid.

G3. Look at the table of roads.

Answer: 8; 9

There are 2 roads from Home to the Park and 4 from the Park to School. Each of the 2 first roads goes with each of the 4 second roads: 4 and 4 make 8 ways through the Park. Add 1 direct road: 9 ways.

Watch for: Adding 2 and 4 to make 6.; Forgetting the direct road.

Investigation (page 5): Every pair that makes 7

The pairs are 1 + 6, 2 + 5 and 3 + 4. After 3 + 4 the pairs repeat (4 + 3 is the same pair). So there are 3 pairs.

Look for: A child who lists 1 + 6, 2 + 5, 3 + 4 in order and says the next pair would be a repeat. Credit any clear reason. A child who counts 0 + 7 has found one more pair; accept it if it is explained.

Exit ticket (page 5)

E1. Count the squares by rows.

Answer: 15 squares

Each row has 5 squares. There are 3 rows. 5 and 5 and 5 make 15.

Watch for: Counting only one row.; Counting a square twice.

E2. Each box is one way to wear a hat and a scarf.

Answer: 9 ways

There are 3 hats and 3 scarves. 3 and 3 and 3 make 9 boxes, so 9 ways.

Watch for: Writing 6 (3 and 3).; Counting only one row.

E3. Four friends, A, B, C and D, make pairs. Here are 5 of the 6 pairs: AB, AC, AD, BC, BD.

Answer: C and D

The 6 pairs are AB, AC, AD, BC, BD and CD. The pair CD is not in the list.

Watch for: Writing DC, which is the same pair (accept it).; Writing a pair that is already in the list.

Sneak-peek puzzles (not in the workbook)

1. 8 squares.
2. 6 ways: each of the 2 hats goes with each of the 3 scarves.